

Shih-Hsin Wang

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ACADEMIC APPOINTMENTS

National Taiwan University

Assistant Professor, Department of Computer Science and Information Engineering

Joint Appointment: Graduate Institute of Networking and Multimedia; AI Center of Research Excellence (AI-CoRE)

Taipei, Taiwan

Aug. 2026 – Present

RESEARCH INTERESTS

- **Generative Models & Test-Time Control:** diffusion models, flow matching, single-step generation, and inference-time steering
- **Geometric Deep Learning & AI for Science:** symmetry-aware learning, geometric graph representations, and molecular/protein modeling

EDUCATION

- **University of Utah**

Ph.D. in Mathematics (Advisors: Bao Wang, Tommaso de Fernex)

Salt Lake City, UT, USA

Aug. 2021 – May 2026

- **National Taiwan University**

B.S. in Mathematics (Honors Program; Presidential Award, 3 times)

Taipei, Taiwan

Sep. 2016 – June 2020

SELECTED HONORS & AWARDS

- **Yushan Young Fellow**, Ministry of Education, Taiwan, 2026
- **ICLR 2025 Oral Presentation**, “A Theoretically-Principled Sparse, Connected, and Rigid Graph Representation of Molecules”

PUBLICATIONS

* indicates equal contribution

Machine Learning **Improving the Efficacy of Test-Time Steering in Masked Diffusion Models with Parallel Tempering**

Lu, P. Y., Lin, H. T., Wang, S. H.

ICML 2026 GenBio Workshop (Spotlight)

Test-Time Guidance for Flow-Based Generative Models via Parallel Tempering on Source Distributions

Wang, S. H.*, Keller, J.*, Transue, T., Brown, D., Strohmer, T., Wang, B.

ICML 2026

RMFlow: Refined Mean Flow by a Noise-Injection Step for Multimodal Generation

Huang, Y., Wang, S. H., Bertozzi, A. L., Wang, B.

ICLR 2026

Plug-and-Play Image Restoration with Flow Matching: A Continuous Viewpoint

Jia, F., Huang, Y., Wang, S. H., Garcia-Cardona, C., Bertozzi, A. L., Wang, B.

SIAM Journal on Imaging Sciences 2026

Towards Multiscale Graph-based Protein Learning with Geometric Secondary Structural Motifs

Wang, S. H., Huang, Y., Transue, T., Baker, J. M., Forstater, J., Strohmer, T., Wang, B.

NeurIPS 2025

Improving Flow Matching by Aligning Flow Divergence

Huang, Y., Transue, T., Wang, S. H., Feldman, W. M., Zhang, H., Wang, B.

ICML 2025

A Theoretically-Principled Sparse, Connected, and Rigid Graph Representation of Molecules

Wang, S. H.*, Huang, Y.*, Baker, J., Sun, Y. E., Tang, Q., Wang, B.

ICLR 2025 (Oral Presentation)

Rethinking Smoothness in Node Features Learned by Graph Convolutional Networks

Wang, S. H.*, Baker, J.*, Hauck, C. D., Wang, B.

Inverse Problems and Imaging, Accepted for Publication

An Explicit Frame Construction for Normalizing 3D Point Clouds

Baker, J.*, Wang, S. H.*, de Fernex, T., Wang, B.

ICML 2024

Rethinking the Benefits of Steerable Features in 3D Equivariant Graph Neural Networks

Wang, S. H., Hsu, Y. C., Baker, J., Bertozzi, A. L., Xin, J., Wang, B.

ICLR 2024

Algebraic Geometry **Arcs on Rational Double Points in Arbitrary Characteristic**

Wang, S. H., de Fernex, T.

Under Review, 2025, arXiv:2508.12423

Families of Jets of Arc Type and Higher (Co)Dimensional Du Val Singularities

Wang, S. H., de Fernex, T.

C.R. Math. Acad. Sci. Paris, Special Volume in Memory of Jean-Pierre Demailly, 2024, arXiv:2306.08291

Other Fields **GenFuzz: GPU-Accelerated Hardware Fuzzing Using Genetic Algorithm with Multiple Inputs**

Lin, D. L., Zhang, Y., Ren, H., Khailany, B., Wang, S. H., Huang, T. W.

ACM/IEEE Design Automation Conference (DAC), 2023

TALKS & PRESENTATIONS

- **NTU Graduate Institute of Networking and Multimedia, 2025:** “Building Geometric Foundations for AI in Molecular Modeling”
- **NTU CSIE, 2025:** “Trustworthy and Efficient AI Through Mathematics – Insights from Geometric Deep Learning and Flow Matching”
- **ICLR 2025 Oral Presentation:** “A Theoretically-Principled Sparse, Connected, and Rigid Graph Representation of Molecules”
- **JMM 2025:** “Expanding the Mathematical Horizons of Machine Learning: Equivariance and Symmetry”
- **SIAM GL 2023:** “Leveraging Geometric Symmetries with Graph Neural Networks”

PRIOR RESEARCH & TEACHING EXPERIENCE

- **Research Assistant** Salt Lake City, UT
Geometric Deep Learning & Generative Models, NSF-supported project, PI: Bao Wang *Jan. 2023 – May 2026*
 - Developed efficient geometric learning and generative-modeling methods, including multiscale protein graphs and canonical-frame constructions for 3D point clouds
- **Visiting Graduate Researcher** Los Angeles, CA
University of California, Los Angeles, Mentor: Andrea Bertozzi *Mar. 2025 – Jun. 2025*
 - Developed RMFlow, an efficient multimodal generative model that refines MeanFlow with a noise-injection step for high-quality one-step generation
- **Research Intern** Los Alamos, NM
Los Alamos National Laboratory, Mentor: Qi Tang *May 2024 – Aug. 2024*
 - Developed a sparse, hyperparameter-free molecular graph representation that preserves rigidity while reducing edge density and improving downstream protein modeling
- **Undergraduate Instructor** Salt Lake City, UT
University of Utah, Department of Mathematics *Aug. 2024 – Dec. 2024*
 - Taught and managed a trigonometry course for 58 undergraduates, including course materials, assessment, grading, and student communication
- **Research Assistant** Salt Lake City, UT
Algebraic Geometry, NSF-supported project, PI: Tommaso de Fernex *May 2022 – Dec. 2022*

ACADEMIC SERVICE

- **Organizer:** two minisymposia on generative models, optimal transport, and inverse problems at SIAM IS26 and MDS26 (Salt Lake City, Nov. 2026)
- Reviewer for top-tier ML conferences: **ICLR 2025–2026, ICML 2024–2026, NeurIPS 2024–2026, AISTATS 2025–2026**
- Reviewer for journals and reviewing venues: **TMLR, SIAM Journal on Applied Algebra and Geometry, ACM Transactions on Sensor Networks**